

From: Scott Feichter scottfeichter@gmail.com
Subject: Rust
Date: July 29, 2025 at 10:18 AM
To: Scott Feichter scottfeichter@gmail.com



Summary - Your Understanding with Corrections:

1. Buffer overflows: Rust prevents them through compile-time checking when possible, runtime checking when necessary, but optimizes away runtime checks when safety can be proven
2. Null pointers: Rust prevents them by using Option enum and not allowing null
3. Data races: Rust prevents them through ownership rules ensuring only one mutable reference at a time per scope
4. Memory leaks: Rust prevents them using ownership that automatically deallocates memory when data goes out of scope



Your core understanding is excellent! These are just minor clarifications on the implementation details.

Scott Feichter
415-702-8512